Learned representations

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Name the target. Trace the encoding. Find the lost distinction.

Audit task-dependent representation collisions and a complete algebraic information repair.

Keep one habit: Complete rectangle trace → scale-matched collision graphic → pictured brightness summary → runnable h-plus-sum repair → independent thresholded encoder.

MAIN LESSON · SECTION 1

What distinctions reach the prediction head?

> A useful representation depends on the target,and constructed features are not automatically learned.

Learning goals

I can trace an **input** through a hidden **representation** and a decision rule.

I can identify distinctions that a **representation** preserves or loses.

I can separate a useful constructed feature from **evidence** of successful learning.

Retrieve the information path and checks

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Debugging shapes and gradients. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Regularisation and early stopping. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Batching and epochs. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Why should a debugging trace inspect scores before ReLU as well as final activations?

> **My starting explanation**
>
> _____
>
> _____
>
> _____
>
> _____

End goal

By the end, I can explain learned **representation**s using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Trace an **input** through a hidden **representation** and a decision rule

Identify distinctions that a **representation** preserves or loses

Separate a useful constructed feature from **evidence** of successful learning

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Declare the task before judging the representation

> Strictly wider has target1;a square has target0.

Complete rectangle target and hand-chosen encoder

Consider fictional rectangles with **input** [width, height] in cm. The task is to identify rectangles whose width is strictly greater than height. The target is 1 for wider rectangles and 0 otherwise.

A square therefore has target 0. This is an arithmetic teaching task, not a claim that a network is needed to measure rectangles.

Define two hidden units: h1 = ReLU(width - height) and h2 = ReLU(height - width). Their weight rows are [1, -1] and [-1, 1], with zero biases. The **representation** h = [h1, h2] records excess width or excess height.

Predict 1 when h1 > 0; otherwise predict 0.

All supplied hidden vectors and predictions

Input	Hidden representation	Prediction
[4, 2]	[2, 0]	1
[2, 4]	[0, 2]	0

Input	Hidden representation	Prediction
[3, 3]	[0, 0]	0
[5, 3]	[2, 0]	1

Separate constructed features from training evidence

These weights were chosen by the lesson author. They demonstrate a **representation** that could be implemented by a network, but no **training** run has established that a model learned it.

Learning would require adjusting parameters using **training** data and recording the procedure.

Name given and missing facts

> **List the given facts and one fact that is not supplied.**

MAIN LESSON · SECTION 3

Trace what survives and inspect a concrete collision

> State both the encoded values and the target distinction at issue.

Head definition and numeric input assumptions

A **prediction** head is the computation that reads the hidden vector and produces an **output**. Keeping it separate lets us inspect what information is available before making a decision.

This example assumes two finite, positive measurements in cm; it is not a general **input** validator.

Complete scores,hidden values,labels and areas

Input[width, height]in cm · Scores[width–height, height–width] · Hidden hafterReLU · Wider prediction · Area in cm²

> **Input[width, height]in cm:** [4, 2]
> **Scores[width–height, height–width]:** [2, −2]
> **Hidden hafterReLU:** [2, 0]
> **Wider prediction:** 1
> **Area in cm²:** 8

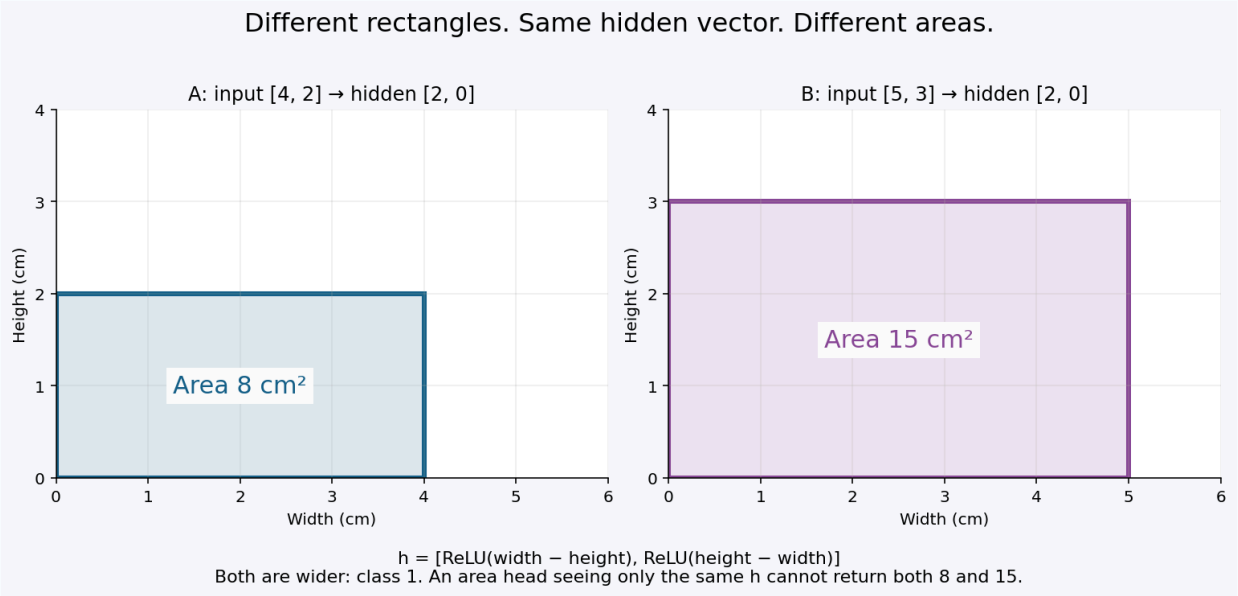
> **Input[width, height]in cm:** [2, 4]
> **Scores[width–height, height–width]:** [−2, 2]
> **Hidden hafterReLU:** [0, 2]
> **Wider prediction:** 0
> **Area in cm²:** 8

> **Input[width, height]in cm:** [3, 3]
> **Scores[width–height, height–width]:** [0, 0]
> **Hidden hafterReLU:** [0, 0]
> **Wider prediction:** 0
> **Area in cm²:** 9

> **Input[width, height]in cm:** [5, 3]
> **Scores[width–height, height–width]:** [2, −2]
> **Hidden hafterReLU:** [2, 0]
> **Wider prediction:** 1

> **Area in cm²:** 15

Scale-matched rectangle collision graphic



Collision meaning and task dependence

A collision means different original inputs share the same h. Whether that matters depends on the target. Rectangles[4, 2]and[5, 3]both should receive wider-label 1, so their collision does not obstruct that comparison. Their required areas 8 and15 differ, so a deterministic area head receiving only h cannot return both exactly. Keeping two hidden components did not preserve the original two measurements.

This is an algebraic property of the supplied encoder, not a measured learned feature or generalization result.

The raw difference width - height can be recovered as h1 - h2. The original width and height cannot in general be recovered: many rectangles share a difference.

Two hidden components do not guarantee preservation of two **input** components.

Runnable encoder and stated comparison head

```
def represent(width, height):
    return [max(0, width - height),
            max(0, height - width)]

def predict(hidden):
    return int(hidden[0] > 0)

examples = [(4, 2), (2, 4), (3, 3), (5, 3)]
for width, height in examples:
    hidden = represent(width, height)
    print([width, height], hidden, predict(hidden))

assert represent(4, 2) == represent(5, 3)
assert predict(represent(3, 3)) == 0
```

Connect each pictured coordinate to its meaning

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Change the head's available information

> A different target can reveal a previously harmless collision.

Current information-loss task

- Give the hidden vectors and areas for [4, 2] and [5, 3]. Explain why increasing only the complexity of a deterministic area head cannot make both exact **prediction**s correct.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured ordered-vector repair

- Change: The head keeps both ordered hidden values instead of their sum.

▼ **Check after predicting**

Prediction: It can distinguish [3, 0] from [0, 3] in this example.

Reason: The **representation** retains the direction of brightness difference when coordinates stay separate.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

See a useful ordered pair collapse into an insufficient summary

> Name the target that the summary can no longer separate.

Complete brightness facts and distinct scope

See when a **representation** preserves a distinction and when it destroys it.

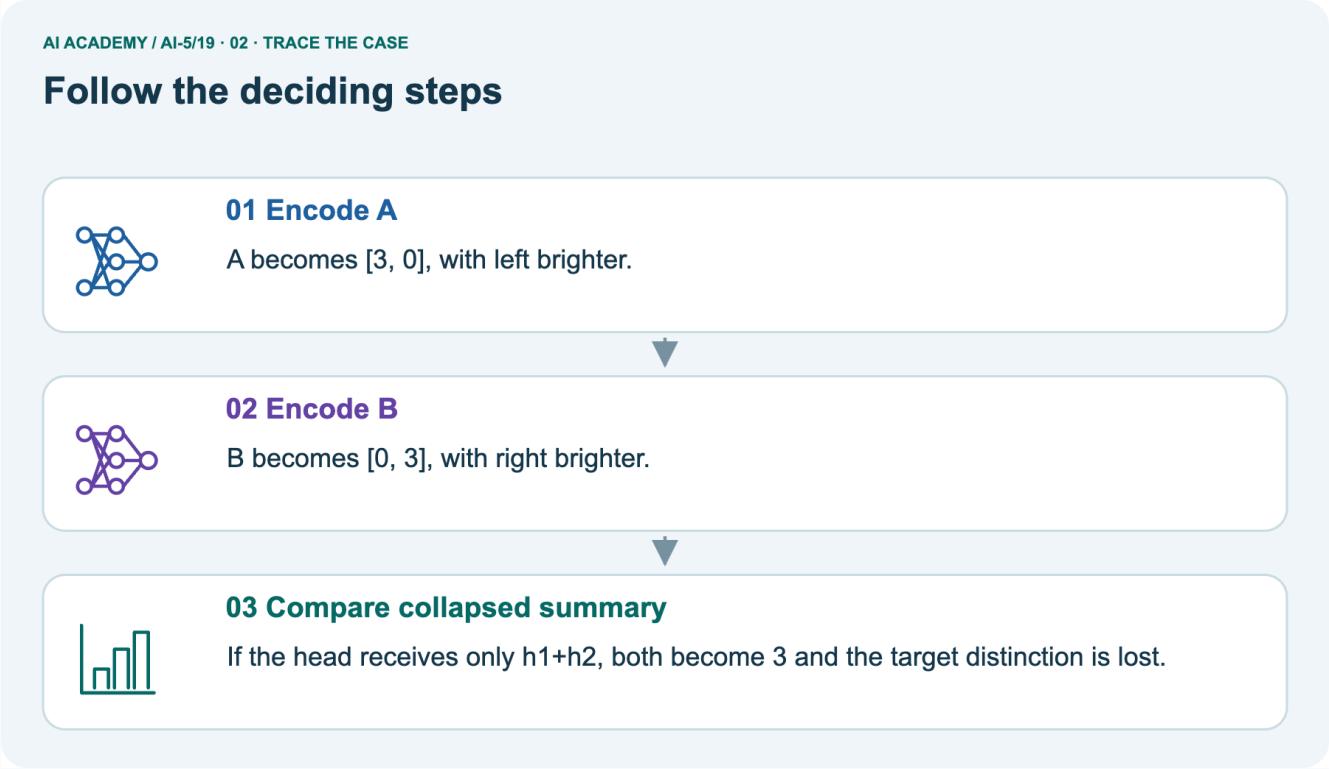
Toy **input** [left, right]; **representation** h=[ReLU(left-right), ReLU(right-left)]. A=[5, 2]; B=[2, 5]; target is left-brighter versus right-brighter.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture uses left/right brightness, not rectangle dimensions. A[5, 2]encodes[3, 0], B[2, 5]encodes[0, 3]. Their ordered vectors preserve which side is brighter. Passing only h1+h2 gives3 for both and loses the target direction, so no deterministic sum-only head can assign both different labels correctly. Keeping both ordered values restores this available distinction for this pair.

These weights are hand-selected; no trained vision model is claimed.

Trace both encodings and collapsed sum



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Encode A	A becomes [3, 0], with left brighter.
2. Encode B	B becomes [0, 3], with right brighter.
3. Compare collapsed summary	If the head receives only h1+h2, both become 3 and the target distinction is lost.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

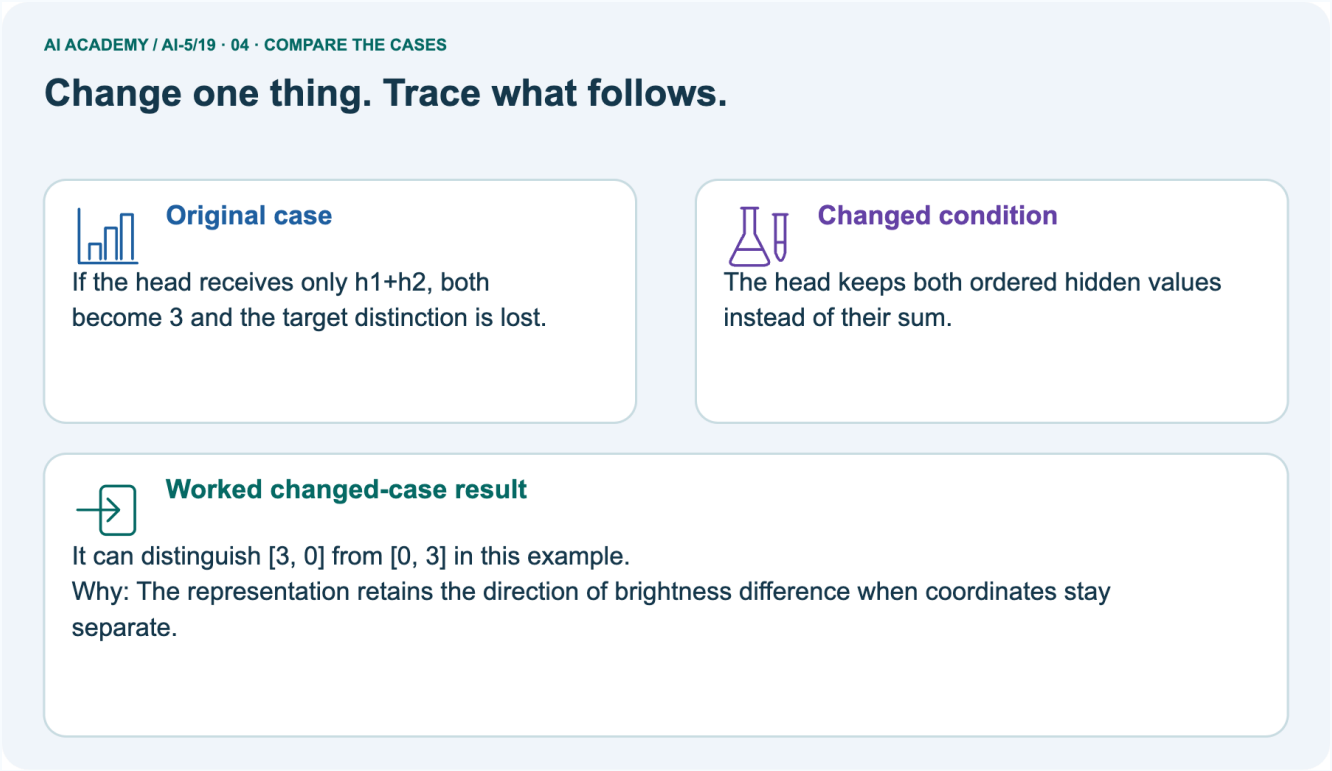
A hidden **representation** decides which distinctions remain available to an **output** head. These two hidden coordinates preserve whether the left or right side is brighter.

Adding them discards that direction and gives both examples the same scalar. A deterministic head cannot assign different labels to identical **input**s; retain the distinction or change the task.

Predict the ordered-pair head

- Change one thing: The head keeps both ordered hidden values instead of their sum. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the deterministic collision and learning limits

- This is a hand-chosen **representation**, not **evidence** that a trained image model actually learned these units.

- Read the labelled picture: Why can no deterministic head using only the sum classify both correctly? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Try the main rectangle area collision

Different rectangles can share a hidden vector

INPUT IN CM	HIDDEN VECTOR	AREA IN SQUARE CM
Width 4 height 2	[2, 0]	8
Width 5 height 3	[2, 0]	15

The representation preserves the wider-than-height answer for these examples but discards information needed to recover their different areas. A deterministic head receiving only this vector must give both examples the same output.

Encode [6, 4] and [7, 5]. Give their wider-than-height predictions and areas. Can a head that sees only h always recover the exact area? Explain using this pair.

▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Learned representations

The facts we will use

AI ACADEMY / AI-5/19

A representation can preserve direction

	left	right	h1	h2
A	5	2	3	0
B	2	5	0	3

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/19

Learned representations

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Encode A

A becomes [3, 0], with left brighter.



2. Encode B

B becomes [0, 3], with right brighter.



3. Compare collapsed summary

If the head receives only h1+h2, both become 3 and the target distinction is lost.

CHANGE ONE THING

The head keeps both ordered hidden values instead of their sum.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Repair missing information rather than only increasing head complexity

> Inspect which distinction each proposed representation preserves.

Task sufficiency,sum collision and algebraic repair

A **representation** can be sufficient for one task and insufficient for another. Our hidden vector supports the wider-than-height rule for every valid **input** under the stated formula.

It fails to identify exact area for all rectangles because a **collision** can contain different areas. No more complicated deterministic head can recover information that is absent from its **input**.

Compare a second encoding s = width + height. Rectangles [4, 2] and [2, 4] both give s = 6 but have different wider-than-height targets. A head receiving only s must fail on at least one of them.

Useful features depend on the question, not just the number of encoded values.

One repair for area is to retain the original measurements. Another is to append s to h. Since d = h1 - h2, width = (s + d) / 2 and height = (s - d) / 2.

This works with exact valid measurements; it does not mean every added feature improves generalisation on noisy data.

Complete h-plus-sum recovery examples

Complete rectangle repair	Information / result
Original h=[excess width, excess height]	Recovers difference d=h1−h2, but not size
Only sum s=width+height	[4, 2]and[2, 4]both s6 but wider targets 1/0; sum alone fails that task
Retain hplus s	width=(s+d)/2, height=(s−d)/2
h[2, 0], s6	d2, width 4, height 2, area 8
Sameh[2, 0], s8	d2, width 5, height 3, area 15

Complete rectangle repair	Information / result
Scope	Exact valid compatible measurements; algebraic recovery is not proof of successful training or noisy-data generalization

Try the sum-only wider-task counterexample

- The alternative **representation** is s = width + height. Use [4, 2] and [2, 4] to decide whether s alone always supports the wider-than-height task.

> **State one repair and the checks you would rerun.**

▼ **Earlier wording, alternatives and picture transcripts**

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise target-dependent reasoning with less help

> Explain the shared representation and whether its required targets agree.

Current task and response

- Encode [8, 3], [3, 8] and [4, 4] with the lesson layer. Apply its **prediction** rule and explain the square boundary.

Encode [6, 4] and [7, 5]. Give their wider-than-height predictions and areas. Can a head that sees only h always recover the exact area? Explain using this pair.

Explain your prediction to a partner before comparing results.

Supported W2record

Use workbook W2 for the connected example “Learned **representation**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual misconception check

- Use the worked case for Learned **representation**s. Proposed explanation: “Two hidden values preserve both **input** measurements”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured sum-only limitation

- Why can no deterministic head using only the sum classify both correctly? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Reserve the independent thresholded encoder

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Prove an information loss

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a complete collision and representation-repair audit

> Every encoder,head,target,input and reconstruction rule is supplied.

Current collision task and runnable original representation

Give the hidden vectors and areas for [4, 2] and [5, 3]. Explain why increasing only the complexity of a deterministic area head cannot make both exact **prediction**s correct.

```
def represent(width, height):
    return [max(0, width - height),
            max(0, height - width)]

def predict(hidden):
    return int(hidden[0] > 0)

examples = [(4, 2), (2, 4), (3, 3), (5, 3)]
for width, height in examples:
    hidden = represent(width, height)
    print([width, height], hidden, predict(hidden))

assert represent(4, 2) == represent(5, 3)
assert predict(represent(3, 3)) == 0
```

Complete runnable reconstruction repair

```
# Hand-constructed representation repair, not a training run.
def represent(width, height):
    return [max(0,width-height), max(0,height-width)]

def recover(hidden, total):
    difference = hidden[0]-hidden[1]
    return (total+difference)/2, (total-difference)/2

for width,height in [(4,2),(5,3),(2,4),(3,3)]:
    hidden = represent(width,height)
    total = width+height
    reconstructed = recover(hidden,total)
    area = reconstructed[0]*reconstructed[1]
    print('input', (width,height), 'h', hidden, 'sum', total,
          'recovered', reconstructed, 'area', area)
    assert reconstructed == (width,height)
print('h alone collides:', represent(4,2)==represent(5,3))
print('sum alone collides:', 4+2==2+4, 'but wider targets differ:', int(4>2),int(2>4))
# Assume exact valid measurements and compatible hidden/total values from this encoder.
```

Normal,boundary,missing,collision and unsupported-learning probes

Complete representation probe	Expected result / action
Normal width 4, height 2	Scores[2, −2], h[2, 0], wider 1, area 8
Boundary square 3, 3	h[0, 0], strict wider 0, area 9
Collision width 5, height 3	Sameh[2, 0], wider 1, area 15; cannot infer exact area fromh
Missing heightNone	Outside finite positive measurement contract; request measurement before calling
Zero or negative rectangle dimension	Invalid for this declared measurement schema; simple encoder does not validate it
Brightness direction A[5, 2]/B[2, 5]	Orderedhseparates; sum 3 collides
Add sum to hfor[4, 2]/[5, 3]	s6/8 allows exact reconstruction 4, 2 and5, 3
Four correct supplied comparison labels	Establishes these forward results, not that weights were learned or that area was predicted
Increase only deterministic area-head complexity	Cannot split identical hwith conflicting required areas; change available information or target specification

Predict,encode,reconstruct and bound the claim

1. Predict every supplied encoding, collision, target and area before running or tracing the original encoder. Then run or trace the complete h-plus-sum repair program and recover the original dimensions. The code assumes valid, exact measurements and compatible h/s values; it is not a parser or general validator. Preserve the original failed area distinction and explain why the repair changes the information available.

   In an actual learned-feature experiment, save training/validation/final roles and a controlled training protocol instead of calling these hand-chosen weights learned.

2. On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Keep the evidence ledger

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: audit a different thresholded representation

> Use its exact formulas and strict wider-than-height target.

Complete new independent encoder and inputs

A different layer uses g = [ReLU(width - height - 1), ReLU(height - width - 1)]. Encode [4, 3] and [3, 4]. State their targets for width strictly greater than height.

Can any deterministic head receiving only g classify both correctly?

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Audit a new representation independently

Attempt this without hints

A different layer uses g = [ReLU(width - height - 1), ReLU(height - width - 1)]. Encode [4, 3] and [3, 4]. State their targets for width strictly greater than height.

Can any deterministic head receiving only g classify both correctly?

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the target and representation evidence together

> The project should show both retained distinctions and recorded collisions.

Project artifact and evidence table

Add a labelled learned **representation**s **evidence** sheet to Image-pattern Network Investigation. Show how you can trace an **input** through a hidden **representation** and a decision rule.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run/trace and limits

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish with one collision and its task-dependent consequence

> Explain the difference between a constructed feature and evidence of learning.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Trace an **input** through a hidden **representation** and a decision rule

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Representation	An encoding of an example used by another computation
Hidden activation	A value produced inside a network after applying an activation function
Collision	Two different inputs receiving the same representation
Learned feature	A feature whose computation depends on parameters adjusted using training data

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Encode [8, 3], [3, 8] and [4, 4] with the lesson layer. Apply its **prediction** rule and explain the square boundary.
- The alternative **representation** is s = width + height. Use [4, 2] and [2, 4] to decide whether s alone always supports the wider-than-height task.
- For h = [2, 0] and s = 8, recover width, height and area. Does this algebra prove that a **training** program learned the **representation** or that it will work well on noisy measurements?

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the representation and its limits

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the new independent thresholded rectangle encoder for the first individual attempt.

Current explanation and response

- In a trained network, hidden features depend on fitted weights and may not have simple human names. A strong **output** result does not by itself show that a unit represents a concept such as shape or fairness.

  Inspect activations on deliberately varied examples, compare with a baseline, and limit conclusions to the **evidence**.

- When testing a learned **representation**, fit pre**process**ing and parameters on **training** data, use validation data for choices, and reserve the final test for the declared final evaluation.

  Compare **representation**s with matched data, head rules and **training** budgets, and report **collision**s and failures. Next week makes the axis and channel meanings of image tensors explicit.

- For h = [2, 0] and s = 8, recover width, height and area. Does this algebra prove that a **training** program learned the **representation** or that it will work well on noisy measurements?

> **A teammate calls the hand-chosen rectangle layer a learned area detector because all four wider-than-height predictions are correct. Identify two problems with this claim and propose one decisive area check.**

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Different rectangles can share a hidden vector

INPUT IN CM	HIDDEN VECTOR	AREA IN SQUARE CM
Width 4 height 2	[2, 0]	8
Width 5 height 3	[2, 0]	15

The representation preserves the wider-than-height answer for these examples but discards information needed to recover their different areas. A deterministic head receiving only this vector must give both examples the same output.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

> **My complete trace of the worked case**

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: See when a **representation** preserves a distinction and when it destroys it.

By the end, I can	How I will show it
Trace an input through a hidden representation and a decision rule	Explain the deciding step, show a trace or test, and state one limit.
Identify distinctions that a representation preserves or loses	Explain the deciding step, show a trace or test, and state one limit.
Separate a useful constructed feature from evidence of successful learning	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- See when a **representation** preserves a distinction and when it destroys it.
- Which labelled step matches this detail? If the head receives only h1+h2, both become 3 and the target distinction is lost.
- 1. Encode B
- 2. Encode A
- 3. Compare collapsed summary
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Compare collapsed summary If the head receives only h1+h2, both become 3 and the target distinction is lost.

A hidden **representation** decides which distinctions remain available to an **output** head. These two hidden coordinates preserve whether the left or right side is brighter.

Adding them discards that direction and gives both examples the same scalar. A deterministic head cannot assign different labels to identical **input**s; retain the distinction or change the task.

Changed-case question and optional hint

- Changed-case practice: The head keeps both ordered hidden values instead of their sum. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. It can distinguish [3, 0] from [0, 3] in this example.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: It can distinguish [3, 0] from [0, 3] in this example. The **representation** retains the direction of brightness difference when coordinates stay separate.

Repair and retain the record

Review Debugging shapes and gradients (ai-5/18). Goal: Check dimension contracts before computing a layer. Short practice: What should a debugging record pair with every gradient value?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Regularisation and early stopping (ai-5/16). Goal: Separate a weight penalty from **prediction** error. Short practice: Why should the penalty value not be reported as validation error?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Batching and epochs (ai-5/13). Goal: Distinguish an example a batch an update and an epoch. Short practice: Why do the single-example orders finish at different weights?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

</details>

<details>
<summary>Reference · Six-picture concept journey</summary>

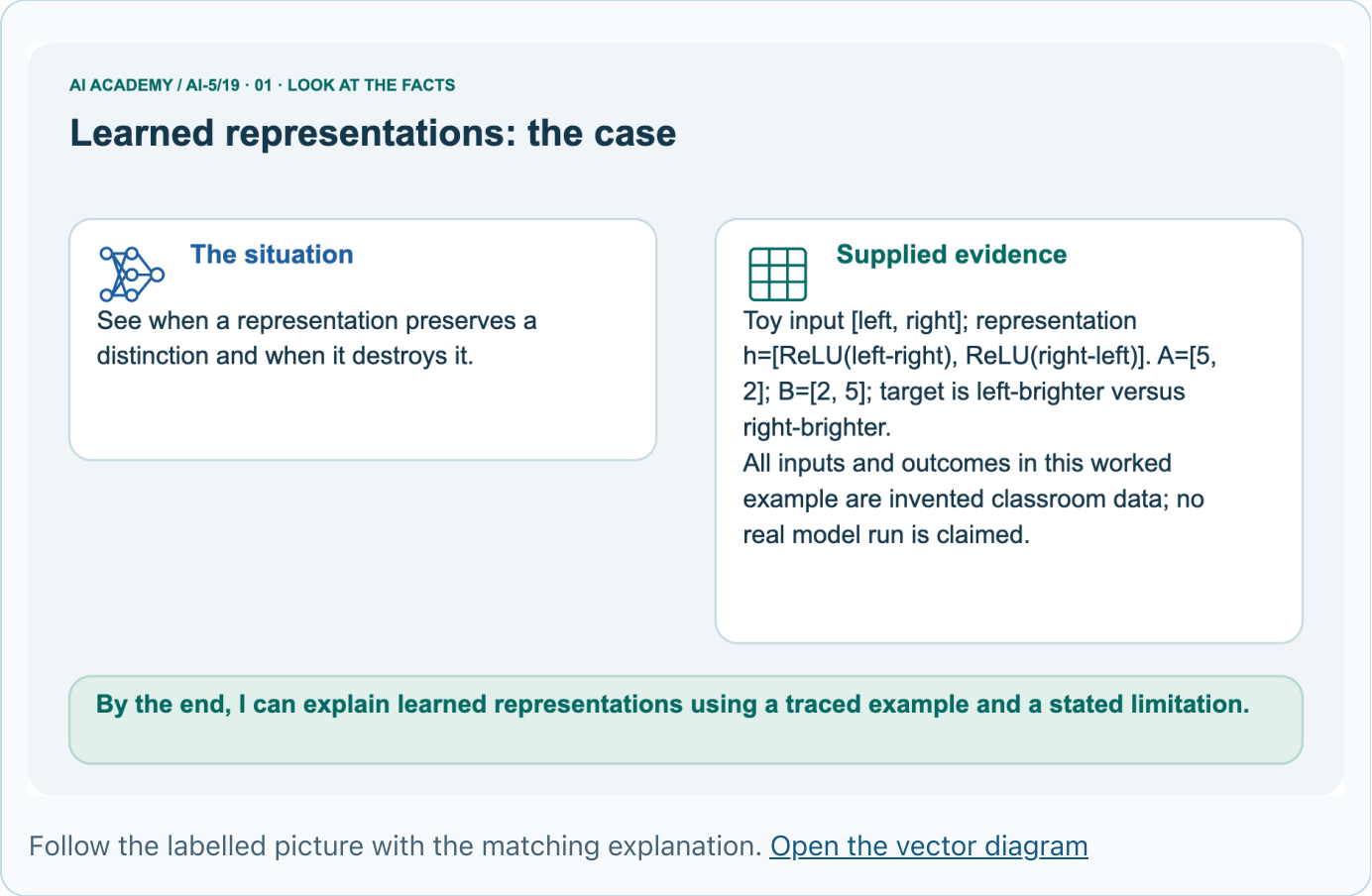
</details>

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Learned representations: the case



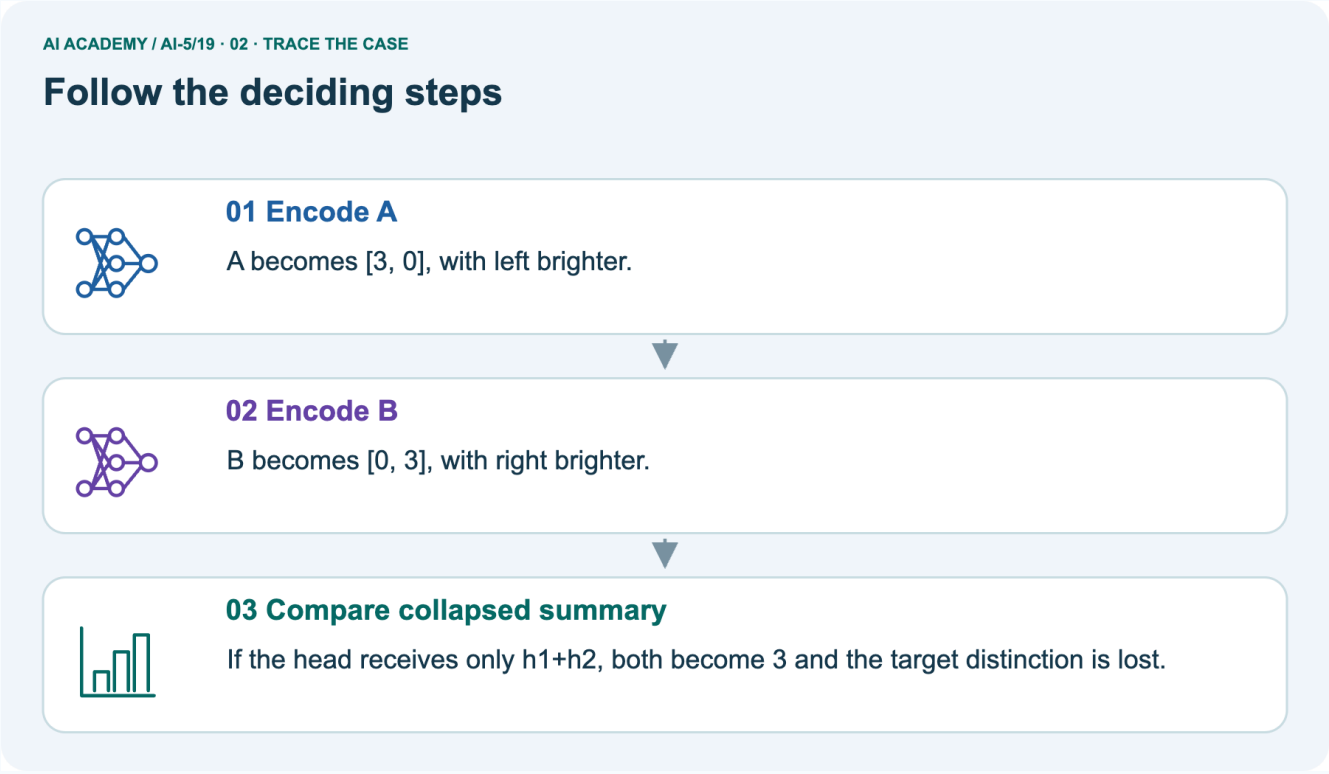
Follow the labelled picture with the matching explanation. Open the vector diagram

See when a **representation** preserves a distinction and when it destroys it. Supplied facts: Toy **input** [left, right]; **representation** h=[ReLU(left-right), ReLU(right-left)].

A=[5, 2]; B=[2, 5]; target is left-brighter versus right-brighter. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

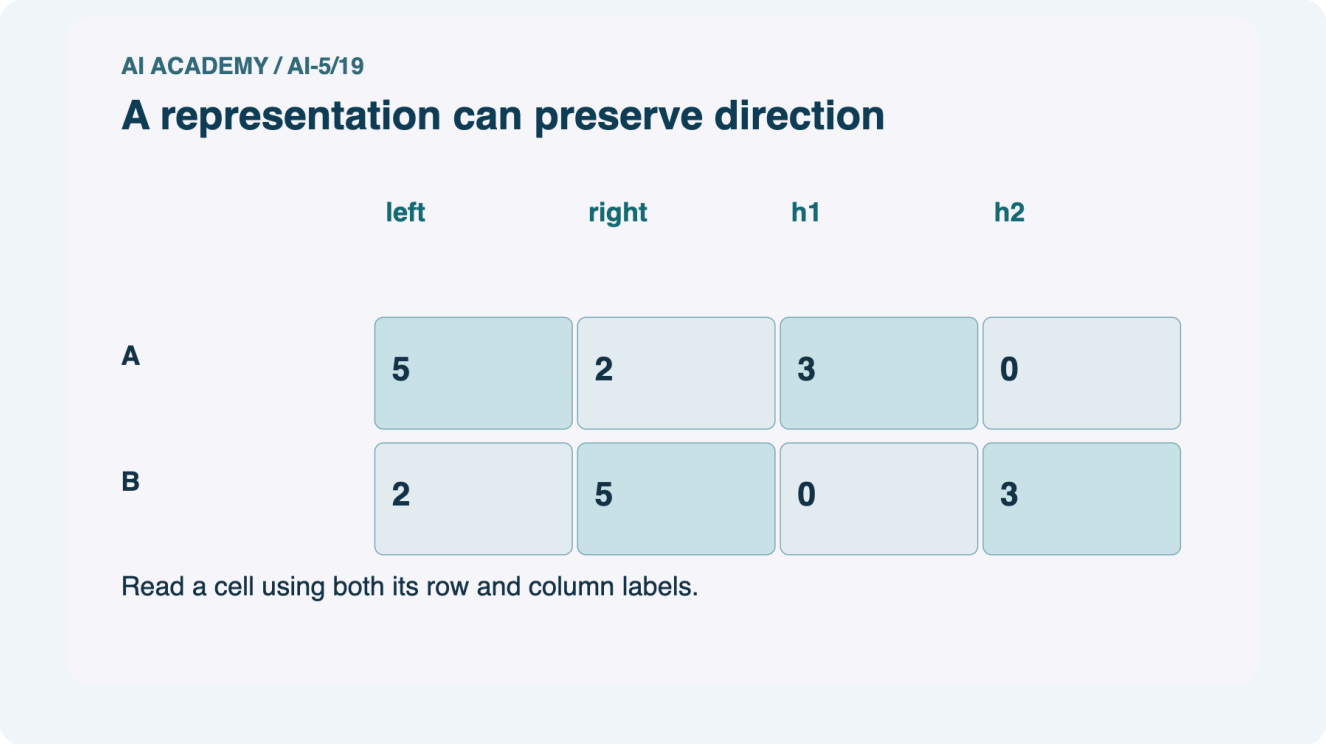


Follow the labelled picture with the matching explanation. Open the vector diagram

Encode A: A becomes [3, 0], with left brighter. → Encode B: B becomes [0, 3], with right brighter. → Compare collapsed summary: If the head receives only h1+h2, both become 3 and the target distinction is lost.

Point to each numbered stage and explain which recorded **input** it uses.

03. A representation can preserve direction

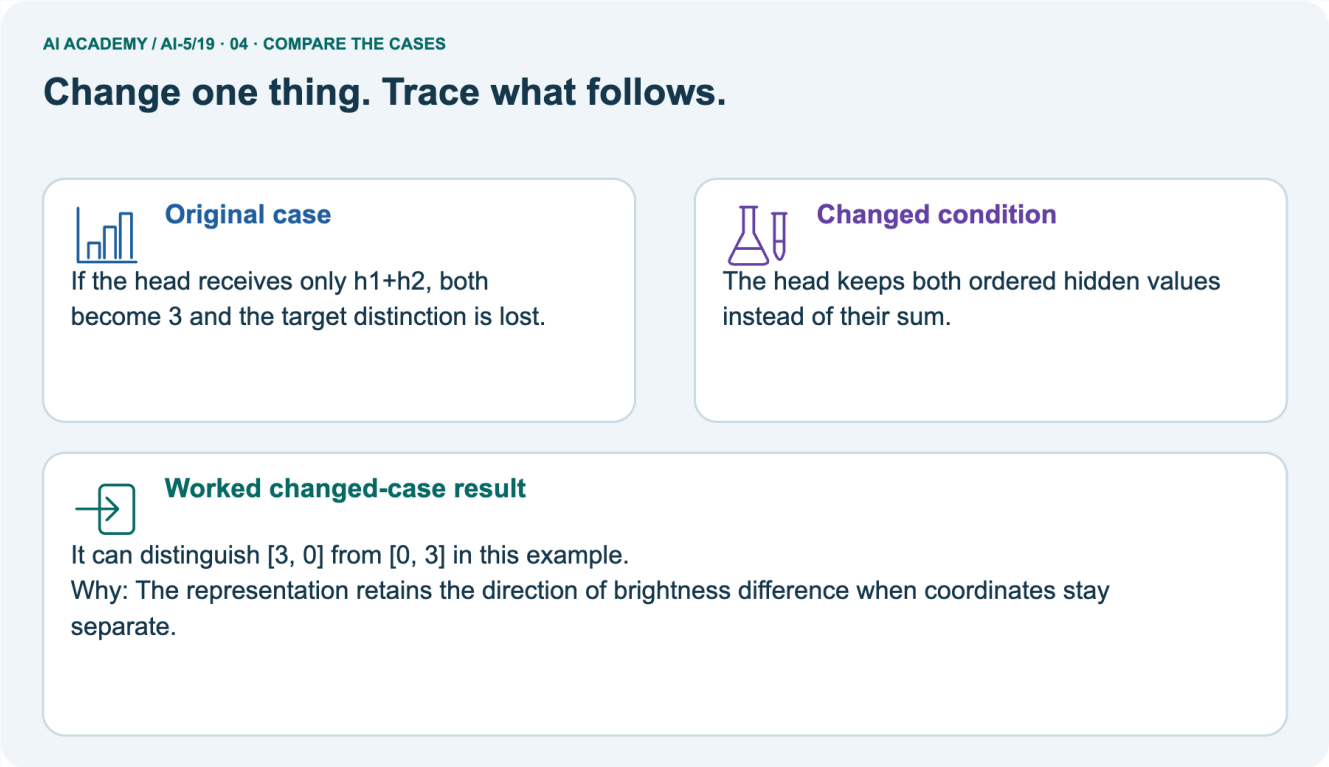


Follow the labelled picture with the matching explanation. Open the vector diagram

A **representation** can preserve direction. A: left = 5, right = 2, h1 = 3, h2 = 0. B: left = 2, right = 5, h1 = 0, h2 = 3.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



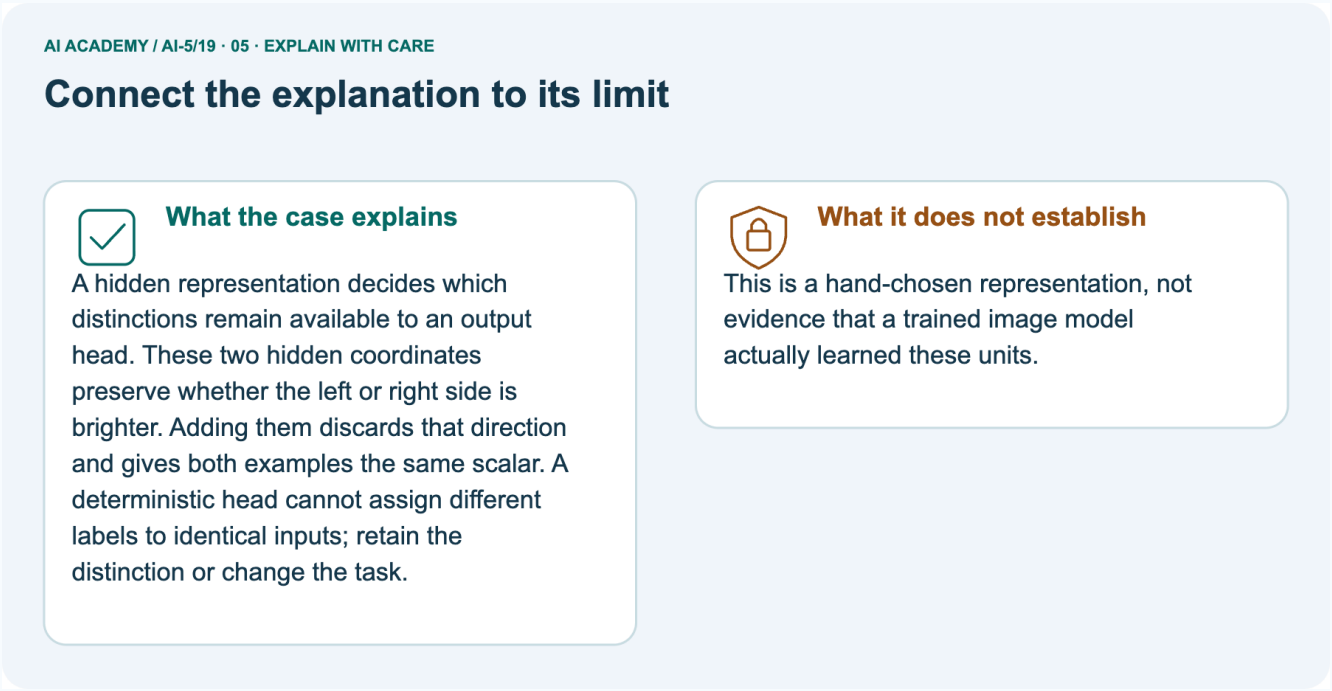
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: If the head receives only h1+h2, both become 3 and the target distinction is lost. Changed condition: The head keeps both ordered hidden values instead of their sum.

Expected result: It can distinguish [3, 0] from [0, 3] in this example. Reason: The **representation** retains the direction of brightness difference when coordinates stay separate.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

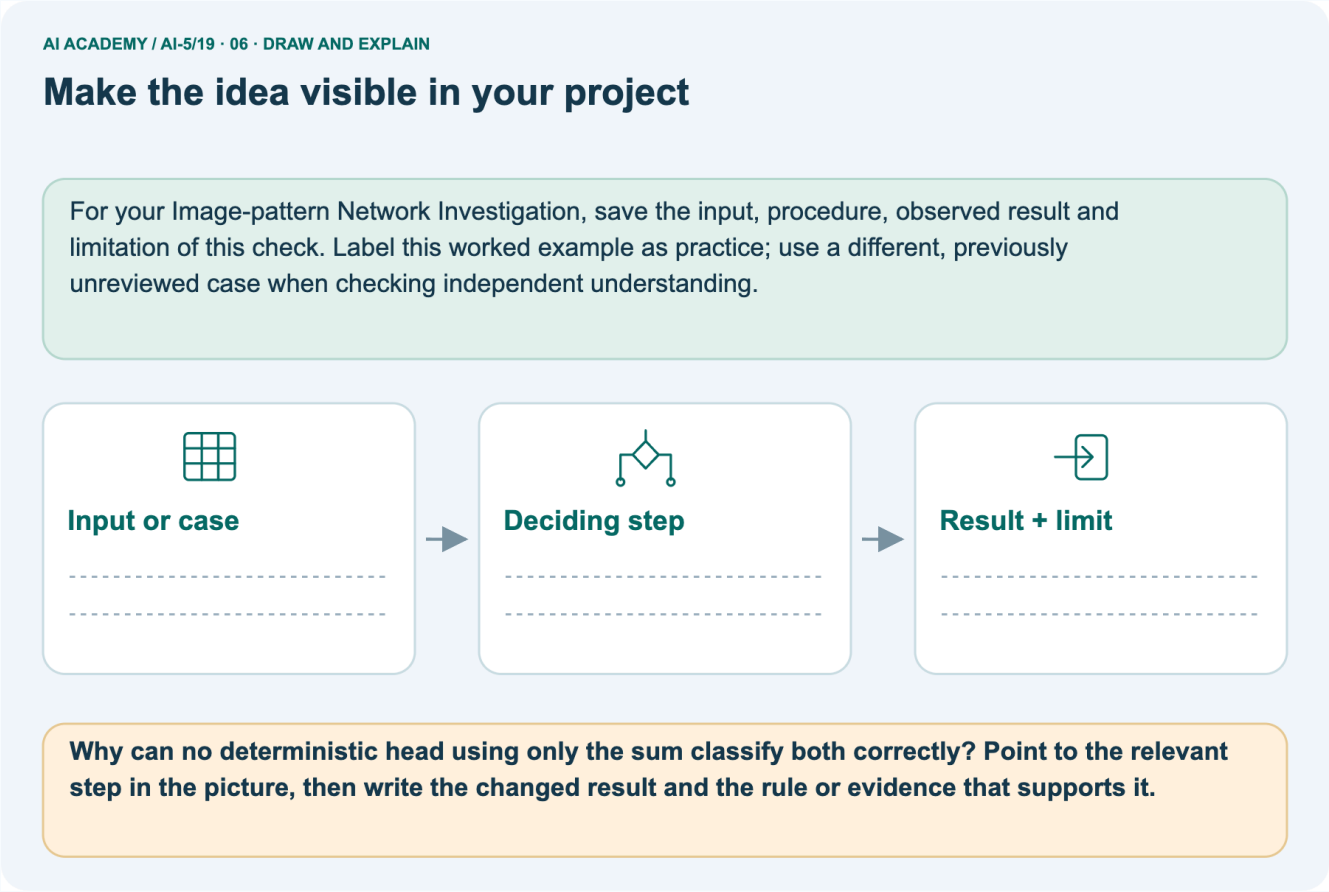
A hidden **representation** decides which distinctions remain available to an **output** head. These two hidden coordinates preserve whether the left or right side is brighter.

Adding them discards that direction and gives both examples the same scalar. A deterministic head cannot assign different labels to identical **input**s; retain the distinction or change the task.

Limit: This is a hand-chosen **representation**, not **evidence** that a trained image model actually learned these units.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why can no deterministic head using only the sum classify both correctly? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Learned representations: the case

02. Follow the deciding steps

03. A representation can preserve direction

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.